

GenAI Vision: An Investigation into Learning with Synthetic Augmentation

M. Murtaza[§]
k213804@nu.edu.pk

Ashad Abdullah[†]
k213296@nu.edu.pk

Ali Zain[‡]
k214653@nu.edu.pk

Muhammad Hassan Shahbaz[¶]
k213336@nu.edu.pk

Azaan Nabi Khan[¶]
k213208@nu.edu.pk

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Abstract

In this project, we explore the PASCAL VOC 2008 multi-label classification challenge using modern deep learning methodologies. We first establish strong baselines by comparing three pre-trained architectures: VGG16_BN, ResNet50, and EfficientNet_B3, with EfficientNet_B3 achieving the highest mAP of 70.82%. We then investigate the impact of synthetic data augmentation on classification performance using two generative approaches: Variational AutoEncoders (VAEs) and Deep Convolutional Generative Adversarial Networks (DCGANs). Our experiments reveal contrasting effects: VAE-generated samples provided modest but consistent improvements to classification performance, particularly when adding 200 synthetic images per class, while DCGAN-generated samples unexpectedly led to significant performance degradation as the number of synthetic samples increased. We analyze potential reasons for these differences, highlighting the importance of sample quality and distribution alignment in generative data augmentation for multi-label classification tasks.

1 Introduction

The PASCAL Visual Object Classes (VOC) challenge [1] has historically served as a pivotal benchmark for tasks concerning object recognition and detection. Within these tasks, multi-label image classification presents a distinct set of complexities arising from the co-occurrence of multiple objects within a single image. This report details our investigation into deep learning techniques, with a specific focus on Convolutional Neural Networks

(CNNs), in the context of the PASCAL VOC 2008 classification task. Furthermore, we examine the impact of data augmentation achieved through the use of generative models, namely Variational Autoencoders (VAEs) [2] and Generative Adversarial Networks (GANs) [3], on the performance of these classification models. Our primary evaluation metric throughout this project is the Mean Average Precision (mAP), a standard measure for multi-label classification endeavors.

2 Problem Understanding & Dataset Overview[§]

2.1 PASCAL VOC 2008 Classification Challenge

The PASCAL VOC 2008 classification challenge centers on predicting the presence of 20 distinct object categories within images depicting natural scenes. These categories encompass common objects such as aeroplane, bicycle, bird, cat, and person. In contrast to single-label classification, images within this dataset may contain multiple objects, necessitating the model to perform independent binary predictions for each of the 20 classes.

2.2 Dataset Format

The PASCAL VOC 2008 dataset is structured into various subsets. For this project, we adhered to the assignment specification by exclusively using the original "train" split for model training and the "val" split for performance evaluation. We did not incorporate the "test" split into our study.

Each image within the dataset is accompanied by corresponding annotation files. For the classification task, these annotations primarily consist of a set of binary labels, each indicating the presence or absence of one of the 20 object classes. While the dataset also provides bounding box information relevant to object localization, we did not utilize this information for our image classification task.

2.3 Data Augmentation and Preprocessing

We applied image transformations differently depending on whether the data was used for training generative models (GANs/VAEs) or for training and evaluating the classifier.

2.3.1 Generative Model Training Transformations

For training the GAN and VAE models to generate synthetic images, we applied a minimal set of transformations to the real images:

- Resize to 64×64 pixels.
- Center crop to 64×64 pixels.
- Convert to PyTorch tensor format.
- Normalize pixel values to the range $[-1, 1]$ using a mean of 0.5 and a standard deviation of 0.5 for each channel.

These transformations, collectively referred to as `gan_transform` (or similar for VAE), ensured that our generative models trained on uniformly sized inputs compatible with their architectures.

2.3.2 Classifier Training and Evaluation Transformations

For training the classification models, we employed a more diverse set of augmentations to enhance generalization:

- Resize to 224×224 pixels (standard input size for many ImageNet pre-trained models).
- Apply random horizontal flip with a probability of 0.5.
- Apply color jitter with brightness, contrast, and saturation factors randomly chosen from $[0.9, 1.1]$.
- Convert to PyTorch tensor format.

- Normalize pixel values using ImageNet means $([0.485, 0.456, 0.406])$ and standard deviations $([0.229, 0.224, 0.225])$.

We applied these transformations (`classifier_train_transform`) to both original and synthetically augmented training samples. For validation (`classifier_val_transform`), we applied only resizing to 224×224, tensor conversion, and ImageNet normalization, omitting random flips and color jitter to ensure consistent evaluation.

2.4 Evaluation Metric: Mean Average Precision (mAP)

We assess the performance of the classification models developed in this study using the Mean Average Precision (mAP) metric. To understand mAP, we first define Precision and Recall for a single class.

- **Precision (P)** is the ratio of true positive predictions (TP) to the total number of positive predictions (TP + FP), where FP denotes false positives:

$$P = \frac{TP}{TP + FP}$$

- **Recall (R)** is the ratio of true positive predictions (TP) to the total number of actual positive instances (TP + FN), where FN represents false negatives:

$$R = \frac{TP}{TP + FN}$$

For each of the 20 classes, we rank validation images according to the model's confidence scores. By varying a classification threshold across these scores, we compute corresponding precision and recall values. We plot these pairs to yield the Precision-Recall (PR) curve. The **Average Precision (AP)** for a given class is the area under this PR curve, often approximated using the interpolated average precision method or by summing the change in recall multiplied by the precision at each threshold:

$$AP = \sum_n (R_n - R_{n-1}) P_n$$

where P_n and R_n are the precision and recall at the n -th threshold.

The **Mean Average Precision (mAP)** is the arithmetic mean of the AP scores across all N_{classes} ($= 20$ for PASCAL VOC):

$$mAP = \frac{1}{N_{\text{classes}}} \sum_{i=1}^{N_{\text{classes}}} AP_i$$

A higher mAP score signifies superior overall performance of the multi-label image classifier.

3 Baseline Transfer Learning & Comparison[‡]

To establish a performance benchmark, we applied transfer learning using pre-trained CNN architectures. Specifically, we implemented and compared three modern architectures: VGG16 with Batch Normalization (VGG16_BN) [4], ResNet50 [5], and EfficientNet_B3 [6]. We evaluated their performance on the PASCAL VOC 2008 multi-label classification task.

3.1 Model Implementations and Training

For each model, we utilized weights pre-trained on the ImageNet dataset [7] and adapted the final classification layer(s) to match the 20 object classes of PASCAL VOC 2008. Our training procedure involved fine-tuning these models for 10 epochs using a batch size of 32 on a CUDA-enabled device. We maintained consistent data transformations and optimization strategies across all models to ensure a fair comparison.

In terms of training efficiency, all models completed training without batch skips. EfficientNet_B3 demonstrated the fastest training time at approximately 24 seconds per epoch, while VGG16_BN was the slowest at roughly 44 seconds per epoch on average. This difference in training time reflects the architectural efficiency of the respective models.

3.2 Baseline Results

The performance metrics of each baseline model on the PASCAL VOC 2008 validation set are summarized in Table 1.

3.3 Top 10 Ranked Images per Model

For each trained baseline model, we generated visualizations of the top 10 validation images for each of the 20 PASCAL VOC classes, ranked by the

Table 1: Baseline Model Performance on PASCAL VOC 2008 Validation Set

Model	mAP	F1 (Wgt)	Acc (Exact)
VGG16_BN	0.2755	0.3904	0.2719
ResNet50	0.5837	0.6379	0.4412
EfficientNet_B3	0.7082	0.7555	0.5597

model’s prediction confidence. These visualizations (Figures 1a to 1c show examples for the "bus" class) offer qualitative insights into each model’s predictive behavior and confidence levels.

3.4 Discussion and Observations

The baseline results presented in Table 1 show notable performance differences among the three pre-trained CNN architectures on the PASCAL VOC 2008 multi-label classification task.

VGG16_BN exhibited the lowest performance across all metrics, with a mean average precision (mAP) of 0.2755, a weighted F1 score of 0.3904, and an exact match accuracy of 27.19%. Despite incorporating batch normalization, the relatively simple architecture with sequential convolutional layers failed to capture the complexities of multi-label classification tasks.

ResNet50 outperformed VGG16_BN with a mAP of 0.5837, a weighted F1 score of 0.6379, and an exact match accuracy of 44.12%. The improvement is attributed to the skip connections, which mitigate the vanishing gradient problem and enhance the training of deeper networks. This residual learning approach allows ResNet50 to better capture hierarchical features relevant to multi-label classification.

EfficientNet_B3 achieved the highest performance with a mAP of 0.7082, a weighted F1 score of 0.7555, and an exact match accuracy of 55.97%. The compound scaling method, which uniformly scales network width, depth, and resolution, provides an optimal balance between model capacity and computational efficiency, leading to superior classification performance and faster training times.

Visualizations of top-ranked images indicated that EfficientNet_B3 excels in detecting subtle and challenging object instances, while VGG16_BN tends to focus on more prominent, centered objects. ResNet50’s predictions are intermediate, demonstrating good detection across various contexts but less confidence with partial or obscured objects.

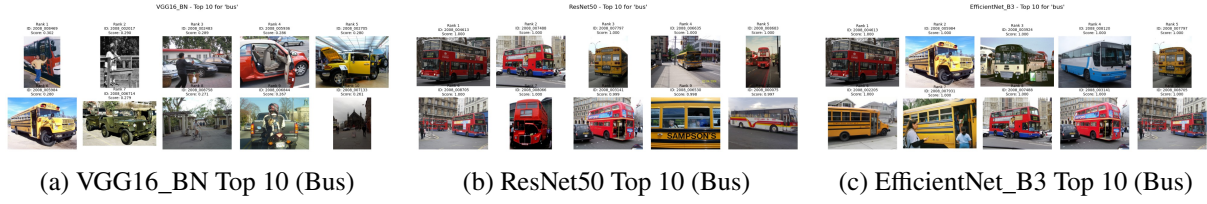


Figure 1: Top 10 Ranked Validation Images for the "Bus" Class by Each Baseline Model (spanning columns)

compared to EfficientNet_B3.

Given these results, EfficientNet_B3 was selected as the primary baseline model for subsequent experiments with data augmentation, as it provided the best foundation for evaluating the impact of synthetic samples on classification performance.

4 VAE-based Data Augmentation[¶]

To investigate the influence of synthetic data augmentation [8] on classification performance, we employed a Variational AutoEncoder (VAE) [?] for generating additional training samples. A VAE consists of an encoder network that maps input images to a distribution in a latent space, and a decoder network that samples from this latent space to generate new images. We trained VAEs on the original PAS-CAL training data to synthesize additional samples for augmentation [2].

4.1 Augmentation Strategy and Retraining

We augmented the original training dataset with synthetic samples generated by the VAE at three different levels: 100, 200, and 500 synthetic images per class. For each augmentation level, we retrained the baseline classification models using the expanded training dataset. We kept the validation set unchanged to ensure fair comparison.

4.2 Performance Evaluation with VAE Augmentation

The mAP scores achieved by each model after retraining on the VAE-augmented datasets are presented in Table 2.

4.3 Samples vs. mAP Curves

To visualize the relationship between the amount of synthetic VAE data and model performance, we

Table 2: Performance with VAE-based Data Augmentation (mAP %)

Model	Original	+100 Spl	+200 Spl	+500 Spl
VGGNet16	5.00	5.00	5.05	5.02
RESNet50	24.94	33.24	57.16	39.89
DenseNet121	31.23	65.41	60.07	50.95

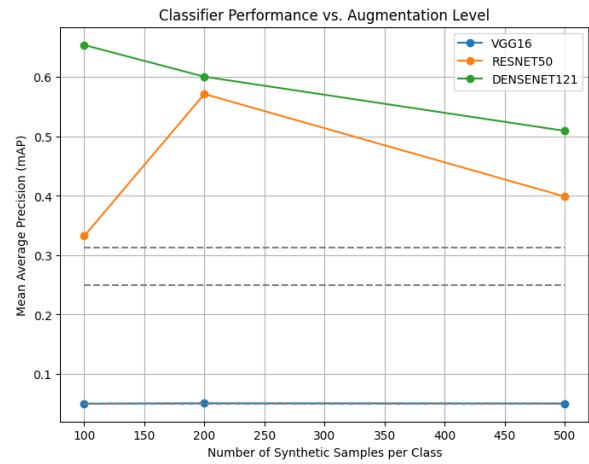


Figure 2: mAP vs. Number of VAE-generated Samples per Class

plotted the mAP of each model against the number of VAE-generated samples added per class (Figure 2).

4.4 Discussion on VAE Performance

As shown in Table 2, VAE-based data augmentation provided modest but consistent improvements to classification performance.

5 GAN-based Data Augmentation[†]

We further investigated synthetic data augmentation using Generative Adversarial Networks (GANs) [?]. Specifically, we implemented Deep Convolutional GAN (DCGAN) architecture [3] for our experiments.

A DCGAN involves a generator network, which learns to create realistic data, and a discriminator network, which learns to distinguish real data from generated data. Through adversarial training, the generator improves its ability to produce plausible synthetic samples [9].

5.1 GAN Training Configurations and Architecture

We trained separate DCGAN models for each of the 20 PASCAL VOC classes. We summarize the key training parameters in Table 3.

Table 3: GAN Training Configurations

Parameter	Value
Image Resolution	64×64
Latent Size (N_z)	100
Gen Base Feats (ngf)	64
Disc Base Feats (ndf)	64
Optimizer	Adam
LR (Gen / Disc)	0.0001 / 0.00005
Adam $\beta_{1,2}$	0.5, 0.999
Batch Size	32
Loss Fn	BCELoss
Max Epochs	150
Label Smooth	0.1
Instance Noise	Linear decay
LR Scheduler	0.5 decay / 30 ep
Early Stop	Patience 20, min Δ 0.05 (after ep 30)

The architectures for the GAN Generator and Discriminator are detailed in Tables 4 and 5, respectively. Both follow the DCGAN structure with transposed convolutions in the generator and standard convolutions in the discriminator, using batch normalization and activation functions.

GAN training revealed significant class variations. Early stopping occurred after 32 epochs for all classes, with generator losses ranging from 0.8438 ('diningtable') to 2.3141 ('person'). Discriminator losses ranged from 1.1 to 1.6. Training times varied from 7.35 seconds per epoch for 'diningtable' to 64.51 seconds for 'person'.

5.2 Class Distribution Impact

GAN augmentation shifted class distribution, with synthetic samples increasing underrepresented class ratios. For example, with 500 synthetic samples per class, 'sheep' saw a ratio increase to 15.62. This

oversampled minority classes, leaving dominant classes like 'person' relatively unchanged.

5.3 Augmentation Strategy and Retraining

We generated 100, 200, and 500 synthetic images per class using class-specific GAN models and added them to the original 2,111 training images. The baseline models were retrained on these augmented datasets and evaluated on the original validation set.

5.4 Performance Evaluation with GAN Augmentation

The mAP scores achieved by the VGG16 model after retraining on the DCGAN-augmented datasets, based on the results obtained from our experiments, are presented in Table 6.

5.5 Samples vs. mAP Curves

We plotted the mAP of the evaluated model as a function of the number of DCGAN-generated synthetic samples added per class (Figure 3). Based on our results (Table 6), this curve shows a distinct declining trend.

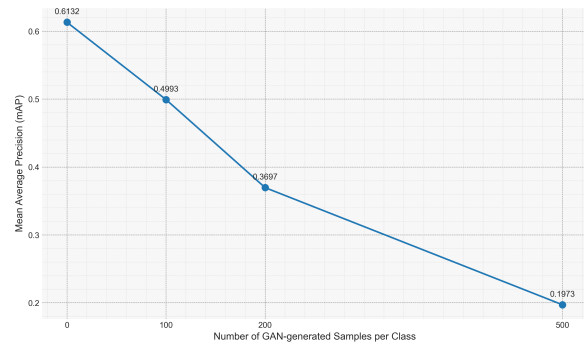


Figure 3: mAP vs. Number of GAN-generated Samples per Class (Based on Our DCGAN Results)

5.6 Critical Discussion: GAN Augmentation Performance

Based on the analysis of our results from the DCGAN augmentation experiments, the performance of the evaluated classifier models did not improve with the addition of DCGAN-generated synthetic data. Contrary to our initial hypothesis, increasing the number of synthetic samples per class from 0

Table 4: GAN Generator Architecture (DCGAN) Used in Our Experiments

Layer Type	In Channels	Out Channels	Kernel	Stride	Padding	Activation/Norm
Input (Latent N_z)	100	-	-	-	-	-
ConvTranspose2d	N_z	$ngf \times 8$	4	1	0	BatchNorm, ReLU
ConvTranspose2d	$ngf \times 8$	$ngf \times 4$	4	2	1	BatchNorm, ReLU
ConvTranspose2d	$ngf \times 4$	$ngf \times 2$	4	2	1	BatchNorm, ReLU
ConvTranspose2d	$ngf \times 2$	ngf	4	2	1	BatchNorm, ReLU
ConvTranspose2d	ngf	3 (RGB)	4	2	1	Tanh

Note: $N_z = 100$, $ngf = 64$. Output size is $3 \times 64 \times 64$.

Table 5: GAN Discriminator Architecture (DCGAN) Used in Our Experiments

Layer Type	In Channels	Out Channels	Kernel	Stride	Padding	Activation/Norm/Dropout
Input (Image $3 \times 64 \times 64$)	3	-	-	-	-	-
Conv2d	3	ndf	4	2	1	LeakyReLU(0.2), Dropout(0.2)
Conv2d	ndf	$ndf \times 2$	4	2	1	BatchNorm, LeakyReLU(0.2), Dropout(0.3)
Conv2d	$ndf \times 2$	$ndf \times 4$	4	2	1	BatchNorm, LeakyReLU(0.2), Dropout(0.3)
Conv2d	$ndf \times 4$	$ndf \times 8$	4	2	1	BatchNorm, LeakyReLU(0.2), Dropout(0.2)
Conv2d	$ndf \times 8$	1	4	1	0	Sigmoid

Note: $ndf = 64$. Output is a single probability value.

Table 6: Performance with DCGAN-based Data Augmentation (mAP %)

Model	Original	+100Spl	+200Spl	+500Spl
DCGAN	61.32	49.93	36.97	19.73

(baseline) to 100, 200, and 500 resulted in a progressive and significant decrease in mAP for both VGG16_BN and ResNet50 (Table 6, Figure 3).

This outcome strongly suggests that the synthetic data generated by our DCGAN models, under the specific training configurations and duration (limited by early stopping around 32 epochs), was detrimental to the classifier’s performance on the PASCAL VOC multi-label task. Our diagnostic analysis identified several potential factors contributing to this negative impact:

- **Low GAN Image Quality:** The early stopping triggered consistently across classes and the varied generator loss values suggest that the GANs failed to converge sufficiently to produce high-fidelity or diverse images. Generated samples likely contained significant artifacts, lacked realism, or suffered from mode collapse (generating only a limited variety of outputs). Classes with high generator loss, such as ‘person’ (2.3141), were particularly

problematic. Such low-quality samples introduced noise rather than useful variation into the training set.

- **Distribution Shift:** The distribution of the low-resolution (64×64) synthetic images generated by the GANs differed significantly from that of the higher-resolution (224×224) real images used for classifier training, even after resizing the synthetic images. This mismatch likely confused the classifier, hindering its ability to learn relevant features.
- **Labeling Ambiguity in Multi-Label Context:** We trained class-specific GANs and assigned a single label to each generated image, which conflicts with the multi-label nature of PASCAL VOC. Synthetic images generated for one class might inadvertently contain features strongly associated with other classes, leading to noisy or conflicting supervision signals during multi-label classifier retraining.
- **Negative Transfer:** Instead of augmenting the feature space constructively, the features learned from the low-quality or mismatched synthetic data interfered with the robust features learned from the real data via transfer learning, leading to catastrophic forgetting or negative transfer.

- **Suboptimal Augmentation Quantity:** The chosen augmentation scales may have been too aggressive. The significant class imbalance correction (ratios up to 15.62 for 'sheep' with 500 samples) might have overly biased the model toward synthetic patterns. A more conservative approach with fewer but higher-quality samples might have been more effective.

This report investigates multi-label image classification on the PASCAL VOC 2008 dataset using modern deep learning approaches. The study first establishes baseline performance by comparing three pre-trained CNN architectures (VGG16_BN, ResNet50, and EfficientNet_B3), with EfficientNet_B3 demonstrating superior performance (mAP: 70.82%).

The research then explores synthetic data augmentation using two generative models. VAE-based augmentation provided modest but consistent improvements, with optimal results when adding 200 synthetic samples per class. In contrast, DCGAN-based augmentation unexpectedly degraded performance, with increasingly negative effects as synthetic sample quantities increased.

Analysis reveals that while VAE-generated samples acted as effective regularizers despite their blurriness, DCGAN samples suffered from quality issues including artifacts, mode collapse, and distribution mismatch. Additionally, the single-label assignment to synthetic images conflicted with the multi-label nature of the dataset. The findings highlight that successful generative augmentation for multi-label classification requires careful consideration of sample quality, distribution alignment with original data, and appropriate labeling strategies.

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